

```
(%i1) load("vect");
(%o1) C:/maxima-5.48.1/share/maxima/5.48.1/share/vector/vect.mac
(%i2) F1(x,y,z):=x^2+3*x*y+y^2;
(%o2) F1(x,y,z):=x^2+3*x*y+y^2
(%i3) grad(F1(x,y,z));
(%o3) grad(y^2+3*x*y+x^2)
(%i4) express(%);
(%o4) 
$$\left[ \frac{d}{dx} (y^2 + 3xy + x^2), \frac{d}{dy} (y^2 + 3xy + x^2), \frac{d}{dz} (y^2 + 3xy + x^2) \right]$$

(%i5) ev(%diff);
(%o5) [3y+2x, 2y+3x, 0]
(%i6) F2(x,y,z):=x^2+y^2+z^2;
(%o6) F2(x,y,z):=x^2+y^2+z^2
(%i7) ev(express(grad(F2(x,y,z))),diff);
(%o7) [2x, 2y, 2z]
(%i8) f1(x,y,z):=[x^2,y^2,z^2];
(%o8) f1(x,y,z):=[x^2,y^2,z^2]
(%i9) div(f1(x,y,z));
(%o9)  $\text{div}\left(\begin{bmatrix} x^2 & y^2 & z^2 \end{bmatrix}\right)$ 
(%i10) express(%);
(%o10)  $\frac{d}{dz} z^2 + \frac{d}{dy} y^2 + \frac{d}{dx} x^2$ 
(%i11) ev(%diff);
(%o11) 2z+2y+2x
(%i12) curl(f1(x,y,z));
(%o12)  $\text{curl}\left(\begin{bmatrix} x^2 & y^2 & z^2 \end{bmatrix}\right)$ 
(%i13) express(%);
(%o13) 
$$\left[ \frac{d}{dy} z^2 - \frac{d}{dz} y^2, \frac{d}{dz} x^2 - \frac{d}{dx} z^2, \frac{d}{dx} y^2 - \frac{d}{dy} x^2 \right]$$

(%i14) ev(%diff);
(%o14) [0,0,0]
(%i15) f2(x,y,z):=[y*z,z*x,x*y];
(%o15) f2(x,y,z):=[y*z,z*x,x*y]
(%i16) ev(express(div(f2(x,y,z))),diff);
(%o16) 0
(%i17) ev(express(curl(f2(x,y,z))),diff);
(%o17) [0,0,0]
```